

Sequential Learning of SAR Image Time Series for Monitoring Earth’s Deformation

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Synthetic Aperture Radar Interferometry (InSAR) has become a cornerstone technique for the precise measurement of Earth’s surface deformation, enabling millimetric monitoring of phenomena such as urban subsidence, volcanic activity, and landslides. By exploiting the phase differences between radar acquisitions performed at different times, InSAR provides highly sensitive deformation measurements over wide areas and under nearly all weather conditions. The advent of systematic satellite missions such as Sentinel-1 has dramatically increased data availability, with free and frequent acquisitions every 6 to 12 days, leading to the generation of dense and high-dimensional SAR time series. While this abundance of data offers unprecedented opportunities for detailed temporal analysis of ground deformation, it also raises major challenges in terms of computational cost, memory requirements, and processing flexibility.

Traditional Multi-Temporal InSAR (MT-InSAR) approaches rely predominantly on so-called offline processing strategies, in which all SAR images in a time series are processed jointly. Although these methods are known for their statistical robustness and estimation accuracy, they suffer from several limitations. First, they exhibit high algorithmic complexity, often scaling cubically with the number of acquisitions. Second, they require significant memory resources to store and manipulate large covariance matrices. Third, they lack flexibility when new acquisitions become available, since integrating additional images typically requires reprocessing the entire time series. These constraints limit their applicability in operational contexts where near-real-time monitoring of continuously evolving deformation phenomena is required.

This work addresses these limitations by introducing a coherent methodological framework for the sequential estimation of interferometric phases in MT-InSAR. The central objective is to design algorithms capable of progressively integrating new SAR acquisitions while preserving the robustness and accuracy of state-of-the-art offline methods, and simultaneously reducing computational complexity and storage requirements. The proposed framework relies on a rigorous statistical modeling of SAR data, temporal correlations, and possible non-Gaussian behavior. In particular, the covariance matrix of the SAR time series is modeled in a factorized form that explicitly separates phase contributions from coherence structure, thereby enabling efficient parameter estimation strategies.

Three complementary sequential approaches are proposed. - The first contribution, termed Sequential Maximum Likelihood Phase Linking (S-MLE-PL), extends classical Phase Linking methods into a sequential setting. Unlike conventional approaches that assume Gaussian-distributed SAR data, S-MLE-PL adopts a more general Scaled Gaussian model, which is better suited to heterogeneous environments and high-resolution imagery, especially in urban areas where heavy-tailed statistics are frequently observed. In this framework, when a new acquisition becomes available, the covariance matrix is updated in a structured block form, allowing the negative log-likelihood (NLL) function to be incrementally updated rather than recomputed from scratch. The joint estimation of the new phase, correlation parameters, variance terms, and scale factors is formulated as a constrained optimization problem. A Block Coordinate Descent (BCD) algorithm is derived, providing closed-form updates for each parameter block. This structured optimization significantly reduces computational complexity, from cubic dependence on the full time series length to quadratic dependence on the number of previously processed images, while maintaining estimation accuracy comparable to that of offline maximum likelihood methods.

- The second contribution, Sequential Covariance Fitting Phase Linking (S-COFI-PL), shifts the perspective from likelihood maximization to covariance matrix fitting. Instead of directly optimizing a statistical likelihood, this approach seeks to minimize a dissimilarity measure between the empirical extended covariance matrix (including new acquisitions) and a theoretical covariance model parameterized by the interferometric phase vector. This formulation naturally accommodates the integration of multiple new acquisitions simultaneously, making it particularly suitable for block-wise updates. The optimization is performed on the complex torus defined by unit-modulus phase vectors, under geometric constraints ensuring physical consistency.

tency. Depending on the structure of the chosen matrix distance, such as the Kullback–Leibler divergence or the Frobenius norm, the optimization is carried out using Majorization–Minimization algorithm. An important strength of this method lies in its flexibility: it can incorporate robust covariance estimators, including phase-only or regularized covariance matrices, thereby enhancing resilience to non-Gaussian disturbances and model mismatches.

- While the two previous methods still require storing historical acquisitions to update the covariance structure, the third contribution introduces a more memory-efficient strategy tailored to very long SAR time series. The proposed Sliding Interferometric Phase Linking (SI-IPL) method relies on a sliding temporal window of fixed size, moving along the time axis with a specified overlap. Within each window, phase estimation is formulated as a covariance fitting problem under a circular complex Gaussian assumption. To ensure temporal consistency between consecutive windows, a penalization term enforces coherence between overlapping phase estimates. The optimization remains constrained to the unit-modulus torus, and the Frobenius norm is favored for its numerical stability and computational efficiency. By limiting the size of the processed covariance matrices, SI-IPL achieves substantial reductions in both computational load and memory usage, while preserving performance close to full sequential methods.

The proposed approaches were validated on both simulated datasets and real SAR time series processed through a complete multi-temporal InSAR chain, including co-registration, interferogram formation, topographic phase correction, phase unwrapping, and line-of-sight displacement estimation. Two contrasted case studies were considered: an urban area affected by significant land subsidence in Mexico City, and a volcanic environment in Hawaii Island. The results demonstrate that the sequential methods achieve mean squared errors and deformation time series estimates comparable to those obtained with reference offline approaches. Furthermore, comparisons with independent GPS measurements, when available, confirm the geophysical consistency of the estimated displacements. From a computational standpoint, the benefits of the proposed sequential framework are substantial. Processing times and memory consumption are significantly reduced, enabling incremental updates of deformation time series as soon as new SAR acquisitions become available.

In summary, this work provides a comprehensive and coherent set of sequential methodologies for interferometric phase estimation in multi-temporal SAR analysis. By combining statistical robustness, optimization on complex manifolds, and efficient incremental updates, the proposed approaches address the growing challenges posed by dense and long SAR time series. They bridge the gap between high-accuracy offline processing and the practical requirements of scalable, near-real-time Earth deformation monitoring.